Supplementary Material

ChiralFold: systematic detection of D-amino acid stereochemistry errors in the Protein Data Bank

Tommaso R. Marena

Contents

1 Supplementary Methods

1.1 Ramachandran benchmark protocol

Era-representative stratified sampling across X-ray ultra (≤1.20 Å), high, medium, low, NMR, and cryo-EM bins (seed 20260612). mmCIF→PDB conversion uses chain-ID remapping (A–Z, 0–9).

Structures with >36 distinct chains skipped. Script: `benchmarks/expand_ramachandran_benchmark.py`.

1.2 Experimental validation protocol

Automated cross-check of all 16 error structures: `benchmarks/experimental_structure_validation.py` queries RCSB entry metadata (method, resolution) and wwPDB CCD InChI stereochemistry; results in `results/experimental_validation_report.json`. Reproduce in Colab: `demos/D_Residue_Experimental_Validation.ipynb`. PDB 5M2K exclusion: `results/5m2k_benchmark_exclusion.json`.

1.3 Mirror-image, AF3 correction, and enumeration

L↔D reflection: $(x, y, z) \to (-x, y, z)$. Because reflection is an isometry, all pairwise distances (and therefore MolProbity-style clashscore) are unchanged before vs. after mirroring (`tests/test_clash_preservation.py`). AF3 correction reflects only the violating Cα (and HA if present) across the N–C–Cβ plane, preserving CA–N, CA–C, and CA–Cβ bond lengths exactly and driving residual chirality violations to 0%. Diastereomer enumeration scores 2^k L/D patterns. Interface scoring uses BSA, H-bonds, and salt bridges.

1.4 Clashscore implementation

ChiralFold clashscore counts non-bonded heavy-atom pairs with van der Waals overlap $\geq$0.4 Å (Bondi radii; KD-tree neighbour search), reported per 1000 heavy atoms. Implementation details that keep scores in the same ballpark as wwPDB/MolProbity on high-quality deposits:

- **First MODEL only** for multi-model (NMR) files.
- **Strip deposited hydrogens**, then place backbone amide H (MolProbity Reduce-style; proline skipped).
- **Exclude covalent 1–2 / 1–3 / 1–4** contacts using standard amino-acid topology (not a brittle distance cutoff), so pairs such as LEU CA–CG are never scored as clashes.
- **Exclude** disulfide SG–SG bridges and donor–acceptor (H-bond) pairs.

Clashscore feeds the composite 0–100 quality score (15 of 100 points) in the CLI/web UI. It is *not* an input to the PDB-wide D-residue survey (signed volume only; independent numpy verifier) nor to the primary Ramachandran calibration ($n = 362$). A secondary 31-structure MolProbity head-to-head panel (`results/molprobity_comparison.json`) was regenerated after these exclusions; mean ChiralFold clashscore on that panel is ∼24 (previously ∼265 from false 1–3 / deposited-H hits). Tests: `tests/test_clash_exclusions.py`, `tests/test_clash_preservation.py`.

1.5 Structure input formats and remote fetch

CLI and Gradio (local / Hugging Face Space) accept:

- Local **PDB** (`.pdb`/`.ent`), **mmCIF** (`.cif`/`.mmcif`; in-core `_atom_site`→PDB conversion with chain-ID remapping; no gemmi required in the core package), and **FASTA** with a UniProt accession in the header (resolved via AlphaFold DB when online).

- Optional network fetch: RCSB by 4-character PDB ID; AlphaFold DB by UniProt accession or `AF-...-Fn` tag (`chiralfold/fetch.py`).

Commands: `chiralfold fetch`, `chiralfold audit --id`, `chiralfold correct-af3 --id`. Fetch is download-only—ChiralFold does not run AF2/Boltz/AF3 inference. The Space prefers bundled CSS and exposes an in-app Light/Dark theme toggle (`?__theme=light|dark`).

1.6 Unusual cases: macrocycles, non-standard ligands, and density

The signed-volume Cα classifier does not require standard protein geometry. Representative edge cases (`demos/Demo_Unusual_Cases_Clash_Safety.ipynb`):

- **Cyclic / strained peptides:** PDB 1XT7 (daptomycin) and 2RMI (astressin).
- **Non-standard CCD ligands:** PDB 1OF6 (eight DTY←L-Tyr); 1BG0/1P52 (DAR←L-Arg).
- **High-resolution density:** PDB 1HHZ (0.99 Å) DAL error.
- **Non-protein Rama exclusion:** PDB 5M2K (vancomycin glycopeptide).

1.7 Five-minute offline reproduction

`pythonbenchmarks/reproduce_d_residue_errors.py` (numpy only; Colab: `demos/Reproduce_PDB_D_Residue_Errors_5min.ipynb`).

1.8 mmCIF known-error re-verification and universe survey

The primary survey used legacy PDB files. We subsequently (i) re-parsed all 16 known-error structures from native RCSB mmCIF and (ii) enumerated the live mmCIF-only universe of entries containing any of the 18 standard D-amino acid CCD codes (RCSB polymer+ligand search, then HEAD filter for missing legacy `.pdb`). The historically cited "∼245" figure counted failed legacy downloads against a broad DPN/DAS/DAL enumeration; the checkable live mmCIF-only set is smaller ($n = 79$ at freeze). Script: `benchmarks/mmcif_d_residue_expansion.py` (`--mode both`); Colab: `demos/Reproduce_mmCIF_D_Residue_Survey.ipynb`. Known-error cohort recovers 29/29 mismatches; the universe cohort reports one additional clear mismatch (9BC4 DLE, $V = +2.82$ Å^3).

1.9 AF3-mimetic correction resource benchmark

Synthetic AF3-mimetic Childs-system peptide PDBs with the published 51% AF3 D-peptide violation rate as external reference. Across DP19, DP9, and DP12 (39 auditable residues; 20 synthetic inverted stereocenters), ChiralFold detected 100% of inverted stereocenters, corrected 100% to the expected signed-volume class, and left 0% residual violations. Full pipeline: 0.0367 s total. Data: `results/af3_resource_benchmark.json`.

1.10 Experimental Childs-system deposited structures

Deposited experimental coordinates for Childs et al. systems (3IWY, 5N8T, 7YH8) were audited as a resource check; this does *not* redistribute AF3 predictions. Clash and composite scores in `results/af3_experimental_systems.json` were regenerated after the topology-aware clashscore fix.

2 Supplementary Notes S1: Lean 4 chirality no-go proofs

2.1 Motivation and scope

ChiralFold audits Cα stereochemistry with the signed tetrahedron volume

$$V = (\mathbf{N} - \mathbf{C}_\alpha) \cdot \big[(\mathbf{C} - \mathbf{C}_\alpha) \times (\mathbf{C}_\beta - \mathbf{C}_\alpha)\big]. \tag{1}$$

Empirically this detects D-label/L-coordinate mismatches. Formally, we ask: *could the same information be recovered from pairwise distances alone?* If not, then distance-only validators and distance-factoring geometric models are structurally blind to enantiomeric sign—motivating an explicit signed-volume check.

We machine-check this obstruction in Lean 4 (v4.28.0) with Mathlib. Production sources contain no `sorry` or `admit`. Build from `formal/chirality_nogo/`:

```
lake build
```

Sources: `ChiralityNoGo.lean`, `OrthogonalChirality.lean`, `ModelAbstraction.lean`, `DistancesOrientVolume.lean`, `Examples.lean`.

What is proved. For four points in $\mathbb{R}^3$, the six pairwise squared distances determine the *square* (hence magnitude) of the oriented tetrahedral volume, but not its *sign*. Any model or classifier whose entire geometric input factors through that distance matrix is invariant on an enantiomeric pair and cannot recover signed chirality on all configurations. Specializing to atom order $(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{C}_\beta)$ yields the protein-facing statement.

What is not proved. This is **not** an impossibility theorem for AlphaFold 3 or for all SE(3)-equivariant neural networks. An SE(3)-equivariant architecture may use angular, tensorial, higher-order, or sequence-derived features and need not factor through a pairwise distance matrix. The formal results state their distance-factoring / full-orthogonal-invariance hypotheses explicitly.

Software scope (not molecular mechanics). ChiralFold does **not** model the complete thermodynamic landscape of structural ensembles, assign Boltzmann weights to conformers, or perform force-field energy minimization. It applies an explicit algebraic data-representation constraint—signed Cα volume and exact coordinate reflection—during mapping and auditing. Structural biologists should not evaluate it as a molecular mechanics or ensemble-dynamics simulator.

2.2 Definitions and notation

Definition 1 (Four-point configuration). *A configuration $P = (a, b, c, d) \in (\mathbb{R}^3)^4$ encodes four labeled points (in the protein case: N, Cα, C, Cβ).*

Definition 2 (Distance matrix). $\mathrm{distMatrix4}(P) \in \mathbb{R}^{4\times 4}$ *has entries $\|P_i - P_j\|^2$ (squared Euclidean distances).*

Definition 3 (Oriented volume).

$$\mathrm{orient4}(P) = \det[b - a \mid c - a \mid d - a]. \tag{2}$$

This is (up to a constant factor) the signed tetrahedron volume used in Eq. 1.

Definition 4 (Factors through distances)**.** *A model* $M : (\mathbb{R}^3)^4 \to \alpha$ factors through distances *if there exists* F *such that* $M(P) = F(\mathrm{distMatrix4}(P))$ *for all* P*.*

Definition 5 (Chirality sign)**.** chiralitySignBool(P) = true *iff* orient4$(P) > 0$ *(Lean Boolean encoding of sign).*

2.3 Informal derivation

Step 1 — Reflection preserves distances, flips orientation. Let R_z denote reflection through the xy-plane, $(x, y, z) \mapsto (x, y, -z)$. For any four points,

$$\mathrm{distMatrix4}(R_z \cdot P) = \mathrm{distMatrix4}(P), \qquad \mathrm{orient4}(R_z \cdot P) = -\mathrm{orient4}(P).$$

(The first holds because R_z is an isometry; the second because $\det(R_z) = -1$.)

Step 2 — Reflection invariance. If $M(P) = F(\mathrm{distMatrix4}(P))$, then

$$M(R_z \cdot P) = F(\mathrm{distMatrix4}(R_z \cdot P)) = F(\mathrm{distMatrix4}(P)) = M(P).$$

Thus M cannot equal orient4 or chiralitySignBool on both P and $R_z \cdot P$ whenever orient4$(P) \neq 0$.

Step 3 — Concrete witness. Take the standard tetrahedron T with vertices $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. Then orient4$(T) = +1$ and orient4$(R_z \cdot T) = -1$, while distMatrix4(T) = distMatrix4$(R_z \cdot T)$. This single witness refutes any claim that distances determine signed orientation.

Step 4 — Gram determinant: magnitude without sign. Let G be the 3×3 Gram matrix of edge vectors from a. Then

$$\mathrm{orient4}(P)^2 = \det(G), \tag{3}$$

and each entry of G is a polynomial in the six pairwise squared distances. Therefore distances determine |orient4| completely, but the two roots $\pm\sqrt{\det(G)}$ remain indistinguishable.

Step 5 — Orthogonal generalization. Every orthogonal map $Q \in \mathrm{O}(3)$ preserves distMatrix4. Oriented volume transforms as orient4$(Q \cdot P) = \det(Q)\,$orient4(P). Hence every *improper* orthogonal map ($\det = -1$) is a chirality-reversing isometry. Any output that factors through distances is invariant under the full orthogonal group and cannot track orient4.

Step 6 — Protein specialization. Instantiating the four points as $(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{C}_\beta)$ yields: the Cα signed-volume surrogate is not a function of those six pairwise distances alone.

2.4 Catalog of machine-checked theorems

Table 1 lists the principal declarations and their consequences. Existing names used in earlier drafts are preserved.

Table 1: Lean 4 declarations (sorry-free) and biological / geometric interpretation.

Lean theorem (file)	Consequence
`same_distances_opposite_orientations_standardTet` (`ChiralityNoGo`)	Standard tetrahedron and its mirror: equal distances, opposite orientation.
`no_distance_only_classifier_separates_standardTet` (`ChiralityNoGo`)	No Boolean function of distances separates that enantiomeric pair.
`improper_orthogonal_distance_chirality_obstruction` (`OrthogonalChirality`)	Every improper orthogonal map preserves distances and flips orientation.
`factorsThroughDistMatrix_orthogonal_invariant` (`ModelAbstraction`)	Every distance-factoring output is O(3)-invariant.
`orient4_not_distance_factorable_via_improper_orthogonal` (`ModelAbstraction`)	Oriented volume cannot factor through distances (improper-orthogonal witness).
`orientTarget_not_factorsThroughDistMatrix` (`ModelAbstraction`)	Abstraction: oriented-volume target is not distance-factorable.
`chiralitySignBool_not_distance_factorable` (`ModelAbstraction`)	Boolean chirality sign is not distance-factorable.
`no_continuous_distance_representation_classifier` (`ModelAbstraction`)	Continuity does not rescue a still distance-factoring representation.
`proteinCaOrient_not_function_of_six_pairwise_distances` (`ModelAbstraction`)	$(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{C}_\beta)$ signed volume needs more than six distances.
`det_gram3_eq_orient_sq` (`DistancesOrientVolume`)	Gram determinant equals squared oriented volume.
`orient_sq_determined_by_pairwise_distances` (`DistancesOrientVolume`)	Distances fix $\lvert\mathrm{orient4}\rvert^2$.
`distance_data_determines_magnitude_not_chirality_sign` (`DistancesOrientVolume`)	Equal distances $\Rightarrow$ equal magnitude, not equal sign.
`standardTetExample_same_distances_different_orientations` (`Examples`)	Concrete normalized example of the ambiguity.

2.5 Concrete example (Examples.lean)

Lean defines

$$\begin{aligned} \texttt{standardTetExample} &= \texttt{standardTet}, \\ \texttt{mirroredStandardTetExample} &= \texttt{reflectZConfig}\,\texttt{standardTetExample}. \end{aligned}$$

Machine-checked facts:

- $\mathrm{distMatrix4}(\texttt{mirrored}) = \mathrm{distMatrix4}(\texttt{standard})$,
- $\mathrm{orient4}(\texttt{standard}) = +1$,
- $\mathrm{orient4}(\texttt{mirrored}) = -1$,
- hence equal distances and unequal signed orientations.

This is the elementary geometric content behind Table 1.

2.6 Proof sketches for headline theorems

`same_distances_opposite_orientations_standardTet`. Compute distMatrix4 and orient4 on `standardTet` and its z-reflection by unfolding definitions and `norm_num`; conclude equality of matrices and inequality of orientations.

`det_gram3_eq_orient_sq`. Expand $\det(G)$ for $G_{ij} = \langle v_i, v_j\rangle$ with $v_1 = b - a$, $v_2 = c - a$, $v_3 = d - a$; identify with $\det[v_1|v_2|v_3]^2 = \mathrm{orient4}^2$ via the classical identity $\det(A^\top A) = \det(A)^2$.

`improper_orthogonal_distance_chirality_obstruction`. For $Q \in \mathrm{O}(3)$ with $\det(Q) = -1$, isometry gives distance preservation; multilinearity of the determinant gives the sign flip of orient4.

`chiralitySignBool_not_distance_factorable`. Assume toward contradiction a factorization through distMatrix4. Apply reflection invariance (Step 2) at the standard-tetrahedron witness (Step 3) to obtain equal Boolean signs despite opposite orientations—contradiction.

`proteinCaOrient_not_function_of_six_pairwise_distances`. Instantiate the four-point abstraction at atom labels $(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{C}_\beta)$ and invoke the general non-factorability of orient4.

2.7 Implications for ChiralFold and AF3

1. **Why signed volume?** It is a minimal non-distance geometric invariant that distinguishes enantiomers.
2. **Why MolProbity's L-only Cα check is insufficient for D-labels.** Checking "standard" L chirality does not audit D-CCD codes; the formal gap is informational (sign), not merely software coverage.
3. **Why AF3's 51% D-peptide rate is consistent with the theory.** Diffusion that respects distance geometry without explicit stereochemical constraints has no access to the missing sign bit—but this does *not* prove that AF3's full architecture is distance-only.
4. **Why post-hoc correction works.** Reflecting Cα across the N–C–Cβ plane injects signed geometry that distances omit, driving residual violations to 0% in the synthetic benchmark (Fig. S4).

3 Supplementary Tables

3.1 Table S1: All 29 D-label/L-coordinate mismatches

Full machine-readable dataset: `results/d_residue_verification.csv` (filter `signed_volume`> 0). Structure-level classification: `results/error_table_verified.csv` and `results/error_classification.json`. Summary: 6 Stereochem, 18 CCD-Code, 3 Mislabel, 2 Borderline across 16 PDB entries.

3.2 Table S2: Formal no-go theorems (compact)

See Table 1 above (Notes S1) for the full catalog. The main text Table 1 highlights five biology-facing theorems.

4 Supplementary Figures

5 Extended statistics

Ramachandran $n = 362$ (post-5M2K exclusion): `results/ramachandran_279struct_chainfix_summary.json` — Spearman $\rho = 0.52$; Pearson $r = 0.85$ (0.33 if 5M2K retained). Pilot $n = 31$ Bland–Altman and bootstrap CIs: `paper/stats_extended.json` via `paper/stats_analysis.py` (seed 42). Deposition-year logistic OR= 1.60/year; CCD clustering $\chi^2 = 113.78$ ($p = 2.33 \times 10^{-16}$).

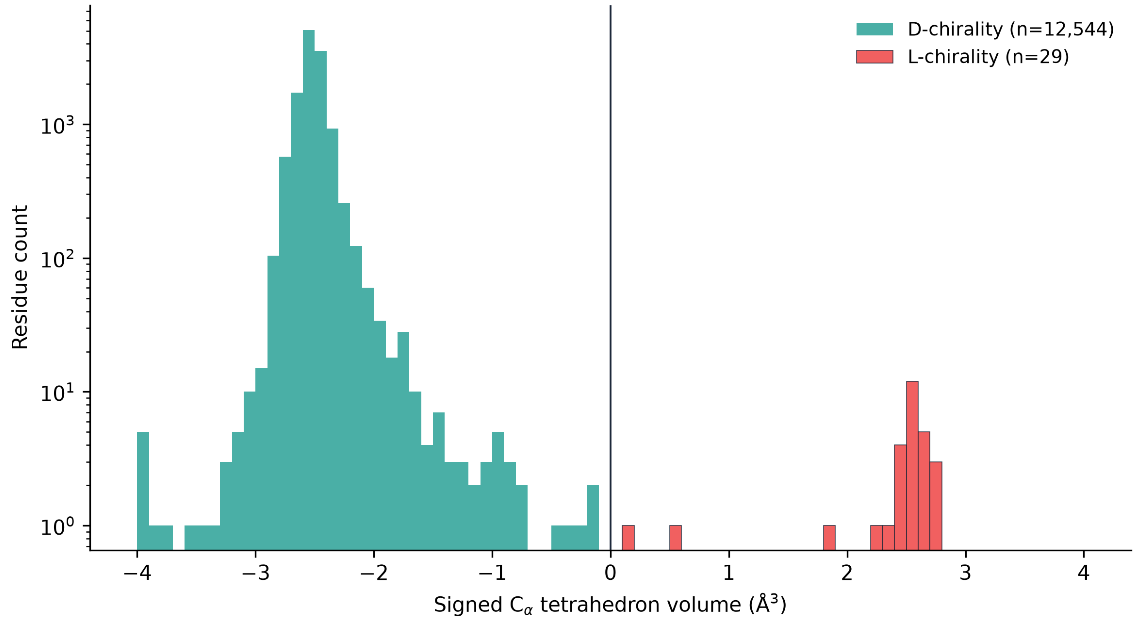

Figure 1: **Figure S1.** Signed Cα tetrahedron volume for 12,573 checkable D-labeled residues (log count). Teal: correct D-chirality ($V < 0$); red: L-coordinates at D-labels ($V > 0$, $n = 29$). Vertical line: $V = 0$ decision boundary used throughout the survey.

Acknowledgments (formal proofs)

Formal verification was assisted by Aristotle (Harmonic). Lean sources and this Notes S1 narrative are part of the ChiralFold v3.5.1 release.

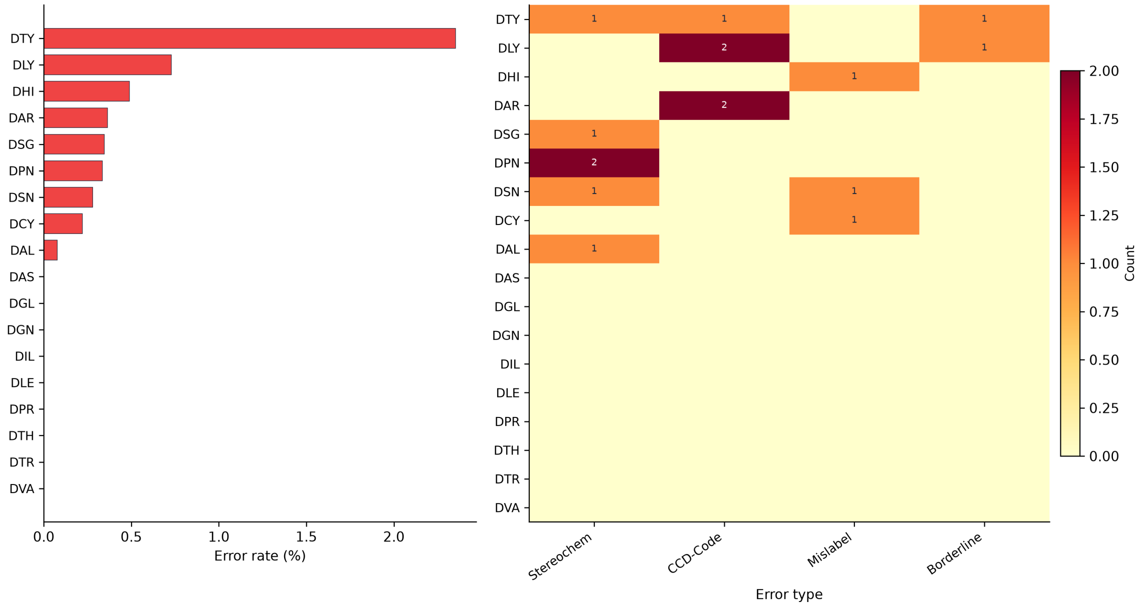

Figure 2: **Figure S2.** Left: per-CCD error rate (%). Right: mismatch counts by error class (Stereochem, CCD-Code, Mislabel, Borderline) across the 18 standard D-amino acid codes. Colorbar: raw count. Supports the claim that errors concentrate in a minority of CCD codes (notably DTY, DLY).

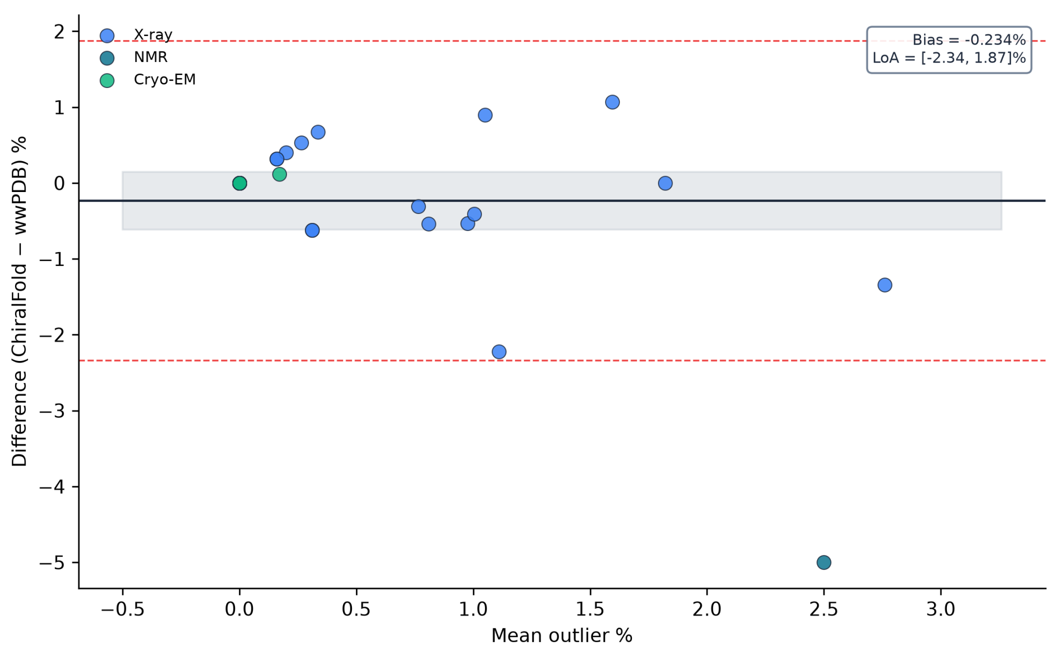

Figure 3: **Figure S3.** Bland–Altman agreement of ChiralFold vs wwPDB Ramachandran outlier percentages on the $n = 31$ pilot benchmark. Bias near zero; dashed lines: 95% limits of agreement. Marker color: experimental method (X-ray / NMR / cryo-EM). Complements the $n = 362$ Spearman calibration in main-text Fig. 2.

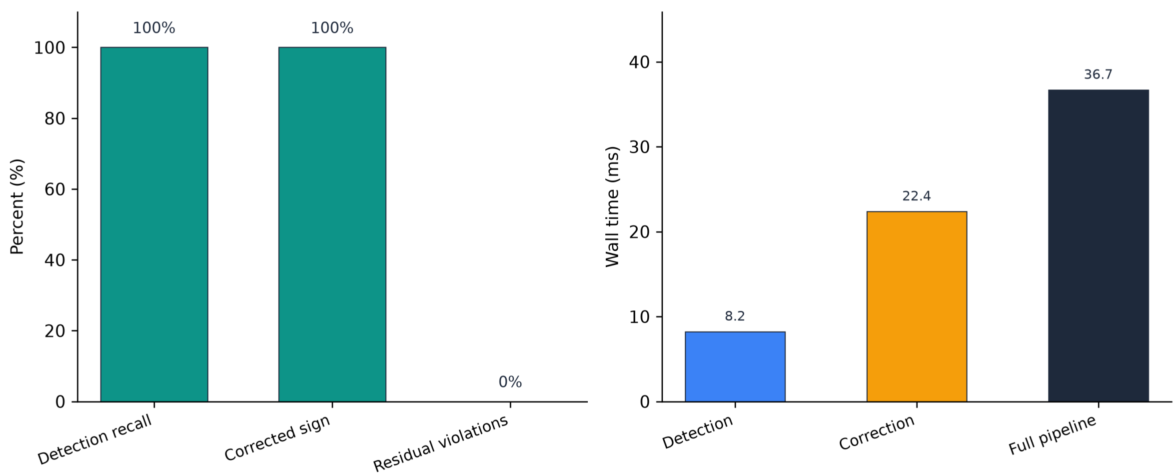

Figure 4: **Figure S4.** Synthetic AF3-mimetic correction benchmark (three Childs-system peptides). Left: detection recall, correction-to-expected-sign, and residual violation rate (%). Right: wall times (ms) for detection, correction, and the full `correct_af3_output` pipeline (36.7 ms total). Supports correction as a fast stereochemistry post-process, not AF3 re-inference.